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Initial clinical experience with a predictive clinical decision support tool for anatomic and reverse total shoulder arthroplasty

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CLINICAL & AGENTIC AI TAKEAWAY

Machine learning-based clinical decision support tools for anatomic and reverse total shoulder arthroplasty show potential for improved patient outcomes and surgeon satisfaction when used responsibly.

ABSTRACT

Purpose: Clinical decision support tools (CDSTs) are software that generate patient-specific assessments that can be used to better inform healthcare provider decision making. Machine learning (ML)-based CDSTs have recently been developed for anatomic (aTSA) and reverse (rTSA) total shoulder arthroplasty to facilitate more data-driven, evidence-based decision making. Using this shoulder CDST as an example, this external validation study provides an overview of how ML-based algorithms are developed and discusses the limitations of these tools. Methods: An external validation for a novel CDST was conducted on 243 patients (120F/123M) who received a personalized prediction prior to surgery and had short-term clinical follow-up from 3 months to 2 years after primary aTSA (n = 43) or rTSA (n = 200). The outcome score and active range of motion predictions were compared to each patient's actual result at each timepoint, with the accuracy quantified by the mean absolute error (MAE). Results: The results of this external validation demonstrate the CDST accuracy to be similar (within 10%) or better than the MAEs from the published internal validation. A few predictive models were observed to have substantially lower MAEs than the internal validation, specifically, Constant (31.6% better), active abduction (22.5% better), global shoulder function (20.0% better), active external rotation (19.0% better), and active forward elevation (16.2% better), which is encouraging; however, the sample size was small. Conclusion: A greater understanding of the limitations of ML-based CDSTs will facilitate more responsible use and build trust and confidence, potentially leading to greater adoption. As CDSTs evolve, we anticipate greater shared decision making between the patient and surgeon with the aim of achieving even better outcomes and greater levels of patient satisfaction. © 2023. The Author(s), under exclusive licence to Springer-Verlag France SAS, part of Springer Nature.

Keywords: Clinical Outcomes, Machine Learning, Rtsa, Atsa, Cdst

Study Overview & Background

Purpose: Clinical decision support tools (CDSTs) are software that generate patient-specific assessments that can be used to better inform healthcare provider decision making. Machine learning (ML)-based CDSTs have recently been developed for anatomic (aTSA) and reverse (rTSA) total shoulder arthroplasty to facilitate more data-driven, evidence-based decision making. Using this shoulder CDST as an example, this external validation study provides an overview of how ML-based algorithms are developed and discusses the limitations of these tools. Methods: An external validation for a novel CDST was conducted on 243 patients (120F/123M) who received a personalized prediction prior to surgery and had short-term clinical follow-up from 3 months to 2 years after primary aTSA (n = 43) or rTSA (n = 200). The outcome score and active range of motion predictions were compared to each patient's actual result at each timepoint, with the accuracy quantified by the mean absolute error (MAE). Results: The results of this external validation demonstrate the CDST accuracy to be similar (within 10%) or better than the MAEs from the published internal validation. A few predictive models were observed to have substantially lower MAEs than the internal validation, specifically, Constant (31.6% better), active abduction (22.5% better), global shoulder function (20.0% better), active external rotation (19.0% better), and active forward elevation (16.2% better), which is encouraging; however, the sample size was small. Conclusion: A greater understanding of the limitations of ML-based CDSTs will facilitate more responsible use and build trust and confidence, potentially leading to greater adoption. As CDSTs evolve, we anticipate greater shared decision making between the patient and surgeon with the aim of achieving even better outcomes and greater levels of patient satisfaction. © 2023. The Author(s), under exclusive licence to Springer-Verlag France SAS, part of Springer Nature.

Methodology & Clinical Focus

Study Classification: other. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Machine learning-based clinical decision support tools for anatomic and reverse total shoulder arthroplasty show potential for improved patient outcomes and surgeon satisfaction when used responsibly.

Extended Repository Abstract

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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